

SUMMARY

- The Laplace transformation converts waveforms in the time domain to transforms in the s domain. The inverse transformation converts transforms into causal waveforms. A transform pair is unique if and only if $f(t)$ is causal.
- The Laplace transforms of basic signals like the step function, exponential, and sinusoid are easily derived from the integral definition. Other transform pairs can be derived using basic signal transforms and the uniqueness, linearity, time integration, time differentiation, and translation properties of the Laplace transformation.
- Proper rational functions with simple poles can be expanded by partial fractions to obtain inverse Laplace transforms. Simple real poles lead to exponential waveforms and simple complex poles to damped sinusoids. Partial-fraction expansions of improper rational functions and functions with multiple poles require special treatment. MATLAB software can be used to find Laplace transforms from $f(t)$ and waveforms from $F(s)$. (see Web Appendix C)
- Using Laplace transforms to find the response of a linear circuit involves transforming the circuit differential equation into the s domain, algebraically solving for the response transform, and performing the inverse transformation to obtain the response waveform.
- The initial and final value properties determine the initial and final values of a waveform $f(t)$ from the value of $sF(s)$ as $s \rightarrow \infty$ and $s \rightarrow 0$, respectively. The initial value property applies if $F(s)$ is a proper rational function. The final value property applies if all of the poles of $sF(s)$ are in the left half plane.

PROBLEMS

OBJECTIVE 9-1 LAPLACE TRANSFORM
(SECTS. 9-1, 9-2, 9-3)

Find the Laplace transform of a given signal waveform using transform properties and pairs, using the integral definition of the Laplace transformation, or software applications. Locate the poles and zeros of the transform and construct a pole-zero diagram.

See Examples 9-1, 9-2, 9-3, 9-4, 9-5, 9-6, 9-7, 9-8, 9-9, 9-10 and Exercises 9-1, 9-2, 9-3, 9-4, 9-5, 9-6, 9-7, 9-8, 9-9, 9-10, 9-11, 9-12, 9-13, 9-14, 9-15.

9-1 Find the Laplace transform of $f(t) = 500[1 - e^{-100t}] u(t)$. Locate the poles and zeros of $F(s)$.

9-2 Find the Laplace transform of $f(t) = 20 \sin(377t) u(t)$. Locate the poles and zeros of $F(s)$.

9-3 Find the Laplace transform of $f(t) = -10 \delta(t) + 10 u(t)$. Locate the poles and zeros of $F(s)$.

9-4 Find the Laplace transform of $f(t) = 20[e^{-200t} - 2e^{-100t}] u(t)$. Locate the poles and zeros of $F(s)$.

9-5 Find the Laplace transform of $f(t) = A[(B + \alpha t) e^{-\alpha t}] u(t)$. Locate the poles and zeros of $F(s)$.

9-6 Find the Laplace transform of $f(t) = 0.005[10 - 10 \cos(1000t)] u(t)$. Locate the poles and zeros of $F(s)$.

9-7 Find the Laplace transform of $f(t) = 5[4 \cos(50t) - 5 \sin(50t)] u(t)$. Locate the poles and zeros of $F(s)$.

9-8 Find the Laplace transform of $f(t) = \delta(t) - 200e^{-20t} \cos(200t) u(t)$. Locate the poles and zeros of $F(s)$.

9-9 Find the Laplace transform of $f(t) = 10[3 - 5t - 2e^{-15t}] u(t)$. Locate the poles and zeros of $F(s)$.

9-10 Find the Laplace transforms of the following waveforms and plot their pole-zero diagrams:

(a) $f_1(t) = [25e^{-15t} - 20e^{-10t}] u(t)$

(b) $f_2(t) = 10[\cos 10t + \cos 20t] u(t)$

9-11 Find the Laplace transforms of the following waveforms and plot their pole-zero diagrams.

(a) $f_1(t) = 2\delta(t) + [200e^{-200t} + 400e^{-400t}] u(t)$

(b) $f_2(t) = [15e^{-200t} + 15 \cos 500t] u(t)$

9-12 9-12 Find the Laplace transforms of the following waveforms. Locate the poles and zeros of $F(s)$. Use MATLAB to verify your results.

(a) $f_1(t) = 5\delta(t) + (625t e^{-25t}) u(t)$

(b) $f_2(t) = [10 + 5e^{-10t}(\cos 10t + \sin 10t)] u(t)$

9-13 9-13 Find the Laplace transforms of the following waveforms. Use MATLAB to verify your results.

(a) $f_1(t) = 5\delta(t - 2)$

(b) $f_2(t) = 10e^{-50(t-1)} u(t - 1)$

(c) $f_3(t) = 20e^{-50(t-10)} u(t - 10)$

9-14 9-14 Use MATLAB to find the Laplace transform of the following waveform

$$f(t) = [10 + 2e^{-10t}] u(t) + [5 \cos 100(t - 0.05)] u(t - 0.05)$$

9-15 Find the Laplace transforms of the following waveforms.

(a) $f_1(t) = \frac{d}{dt} (50e^{-1000t} \cos 200kt) u(t)$

(b) $f_2(t) = \int_0^t 20e^{-10x} dx + 10u(t) + 20 \frac{de^{-10t}}{dt} u(t)$

9-16 Consider the waveform in Figure P9-16.

(a) Write an expression for the waveform $f(t)$ using step and ramp functions.

- (b) Use the time-domain translation property to find the Laplace transform of the waveform $f(t)$ found in part (a).
 (c) Verify the Laplace transform found in (b) by applying the definition of the Laplace transformation in Eq. (9-2) to the waveform $f(t)$ found in (a).

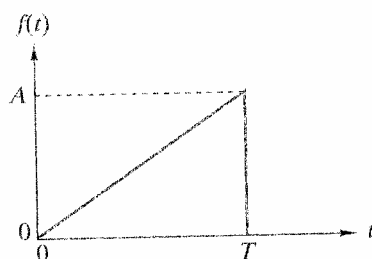


FIGURE P9-16

9-17 Consider the waveform in Figure P9-17.

- (a) Write an expression for the waveform $f(t)$ in Figure P9-17 using step functions.
 (b) Use the time-domain translation property to find the Laplace transform of the waveform $f(t)$ found in part (a).
 (c) Verify the Laplace transform found in (b) by applying the definition of the Laplace transformation in Eq. (9-2) to the waveform $f(t)$ found in (a).

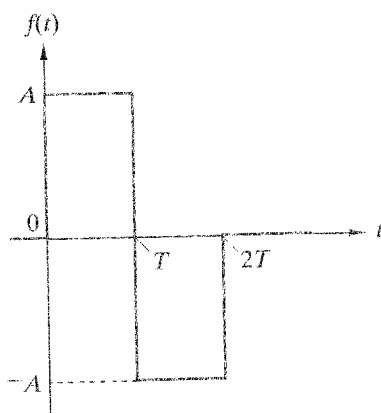


FIGURE P9-17

9-18 For the following waveform: $f(t) = [500 + 100e^{-500t} \sin 1000t]u(t)$

- (a) Find the Laplace transform of the waveform. Locate the poles and zeros of $F(s)$.
 (b) Validate your result using MATLAB.
 9-19 Consider the waveform in Figure P9-19.
 (a) Write an expression for the waveform $f(t)$ in Figure P9-19 using a delayed exponential.
 (b) Use the time-domain translation property to find the Laplace transform of the waveform $f(t)$ found in part (a).
 (c) Verify the Laplace transform found in (b) by applying the definition of the Laplace transformation in Eq. (9-2) to the waveform $f(t)$ found in (a).

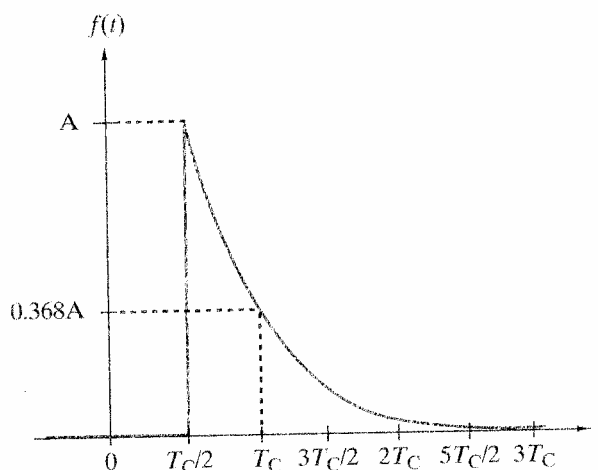


FIGURE P9-19

OBJECTIVE 9-2 INVERSE TRANSFORMS (SECT. 9-4)

- (a) Find the inverse transform of a given Laplace transform using partial fraction expansion, basic transform properties and pairs, or using software tools.
 (b) Given a pole-zero diagram, find the respective transform. See Examples 9-11, 9-12, 9-13, 9-14, 9-15 and Exercises 9-16, 9-17, 9-18, 9-19, 9-20, 9-21, 9-22, 9-23, 9-24, 9-25.

9-20 Find the inverse Laplace transforms of the following functions:

(a) $F_1(s) = \frac{50}{s(s+50)}$

(b) $F_2(s) = \frac{s+1}{(s+2)(s+3)}$

9-21 Find the inverse Laplace transforms of the following functions:

(a) $F_1(s) = \frac{s+30}{s(s+40)}$

(b) $F_2(s) = \frac{(s+10)(s+20)}{s(s+50)(s+100)}$

9-22 Find the inverse Laplace transforms of the following functions:

(a) $F_1(s) = \frac{5000(s+1000)}{(s+500)(s+5000)}$

(b) $F_2(s) = \frac{5s^2}{(s+100)(s+500)}$

9-23 Find the inverse Laplace transforms of the following functions:

(a) $F_1(s) = \frac{900}{(s+10)^2 + 30^2}$

(b) $F_2(s) = \frac{3(s+10)}{(s+10)^2 + 30^2}$

9-24 Find the inverse Laplace transforms of the following functions and sketch their waveforms for $\beta > 0$:

(a) $F_1(s) = \frac{\beta(s + \beta)}{s(s^2 + \beta^2)}$

(b) $F_2(s) = \frac{s(s + \beta)}{s^2 + \beta^2}$

9-25 Find the inverse Laplace transforms of the following functions:

(a) $F_1(s) = \frac{\alpha^2}{s^2(s + \alpha)}$

(b) $F_2(s) = \frac{\alpha^2}{s(s + \alpha)^2}$

9-26 Find the inverse Laplace transforms of the following functions:

(a) $F_1(s) = \frac{600}{(s + 10)(s + 20)(s + 30)}$

(b) $F_2(s) = \frac{2(s + 10)}{(s + 15)(s + 20)}$

9-27 Find the inverse Laplace transforms of the following functions then validate your answers using MATLAB:

(a) $F_1(s) = \frac{16s}{(s + 3)(s^2 + 11s + 10)}$

(b) $F_2(s) = \frac{5(s^2 + 9)}{s(s^2 + 25)}$

9-28 Find the inverse transforms of the following functions:

(a) $F_1(s) = \frac{(s + 10000)(s + 100000)}{s(s + 1000)(s + 50000)}$

(b) $F_2(s) = \frac{3(s^4 + 10s^2 + 4)}{s(s^2 + 1)(s^2 + 4)}$

9-29 Find the inverse transforms of the following functions:

(a) $F_1(s) = \frac{300(s + 50)}{s(s^2 + 40s + 300)}$

(b) $F_2(s) = \frac{1000s}{(s + 5)(s^2 + 4s + 8)}$

9-30 Find the inverse transforms of the following functions:

(a) $F_1(s) = \frac{16(s^2 + 256)}{s(s^2 + 8s + 32)}$

(b) $F_2(s) = \frac{3(s^2 + 20s + 400)}{s(s^2 + 50s + 400)}$

9-31 Find the inverse Laplace transforms of the following functions then validate your answers using MATLAB:

(a) $F_1(s) = \frac{(s + 100)^2}{(s + 50)^2(s + 200)}$

(b) $F_2(s) = \frac{(s + 50)^2}{(s + 100)^2(s + 200)}$

9-32 A certain transform has a simple pole at $s = -20$, a simple zero at $s = -\gamma$, and a scale factor of $K = 1$. Select values for γ so the inverse transform is

(a) $f(t) = \delta(t) - 5e^{-20t}$ (b) $f(t) = \delta(t)$ (c) $f(t) = \delta(t) + 5e^{-20t}$

9-33 Find the inverse transforms of the following functions:

(a) $F_1(s) = \frac{s(s + 10)(s + 20)}{(s + 5)(s + 30)(s + 50)}$

(b) $F_2(s) = \frac{(s + 1000)(s + 5000)}{(s + 2000)}$

9-34 Find the inverse transforms of the following functions:

(a) $F_1(s) = \frac{s^2}{(s + 5)}$

(b) $F_2(s) = \frac{(s + 1000)^2}{(s + 2000)^2}$

9-35 Find the inverse transforms of the following functions:

(a) $F_1(s) = \frac{e^{-5s}(s + 20)}{(s + 10)(s + 30)}$

(b) $F_2(s) = \frac{se^{-5s} + 20}{(s + 10)(s + 30)}$

(c) $F_3(s) = \frac{s + 20e^{-5s}}{(s + 10)(s + 30)}$

9-36 Use MATLAB to find the inverse transform and plot the poles and zeros of the following function:

$$F(s) = \frac{300s(s^2 + 30s + 400)}{(s + 20)(s^3 + 6s^2 + 16s + 16)}$$

9-37 Use MATLAB to find the inverse transform and plot the poles and zeros of the following function:

$$F(s) = \frac{500(s^3 + 2s^2 + s + 2)}{s(s^3 + 4s^2 + 4s + 16)}$$

9-38 Find the transform $F(s)$ from the pole-zero diagram of Figure P9-38. K is 5.

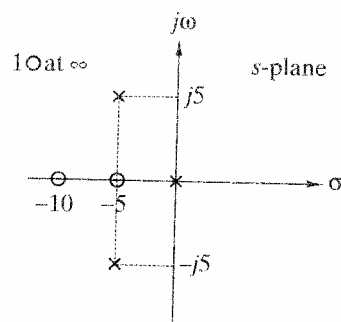


FIGURE P9-38

9-39 Find the transform $F(s)$ from the pole-zero diagram of Figure P9-39. K is 5×10^6 .

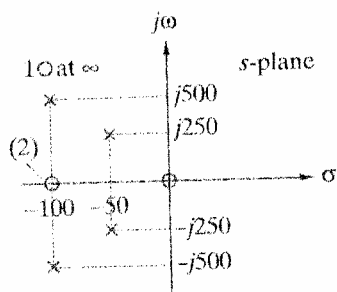


FIGURE P9-39

9-40 Find the transform $F(s)$ from the pole-zero diagram of Figure P9-40. K is 50.

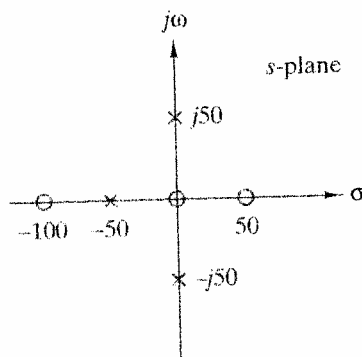


FIGURE P9-40

OBJECTIVE 9-3 CIRCUIT RESPONSE USING LAPLACE TRANSFORMS (SECT. 9-5)

Given a first- or second-order circuit:

- Determine the circuit differential equation and the initial conditions (if not given).
- Transform the differential equation into the s domain and solve for the response transform.
- Use the inverse transformation to find the response waveform.

See Examples 9-16, 9-17, 9-18, 9-19 and Exercises 9-26, 9-27, 9-28, 9-29, 9-30.

9-41 Use the Laplace transformation to find the $v(t)$ that satisfies the following first-order differential equations:

- $250 \frac{dv(t)}{dt} + 2500v(t) = 0$, $v(0^-) = 50$ V
- $\frac{dv(t)}{dt} + 300v(t) = 600u(t)$, $v(0^-) = -150$ V

9-42 Use the Laplace transformation to find the $i(t)$ that satisfies the following first-order differential equation:

$$\frac{di(t)}{dt} + 500i(t) = [0.100e^{-100t}]u(t), \quad i(0^-) = 0$$
 A

9-43 The switch in Figure P9-43 has been open for a long time and is closed at $t = 0$. The circuit parameters are $R = 1$ k Ω , $L = 100$ mH, and $V_A = 15$ V.

- Find the differential equation for the inductor current $i_L(t)$ and initial condition $i_L(0)$.
- Solve for $i_L(t)$ using the Laplace transformation.

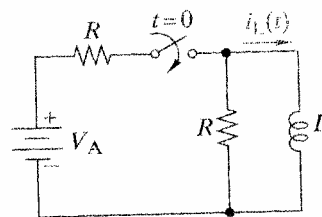


FIGURE P9-43

9-44 The switch in Figure P9-43 has been closed for a long time and is opened at $t = 0$. The circuit parameters are $R = 50$ Ω , $L = 200$ mH, and $V_A = 50$ V.

- Find the differential equation for the inductor current $i_L(t)$ and initial condition $i_L(0)$.
- Solve for $i_L(t)$ using the Laplace transformation.

9-45 The switch in Figure P9-45 has been open for a long time. At $t = 0$ the switch is closed.

- Find the differential equation for the capacitor voltage and initial condition $v_C(0)$.
- Find $v_O(t)$ using the Laplace transformation for $v_S(t) = 25u(t)$ V.

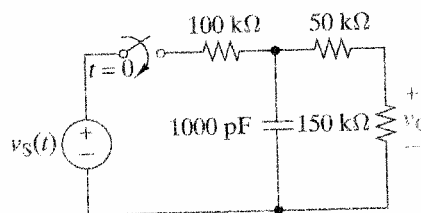


FIGURE P9-45

9-46 Repeat Problem 9-45 for the input waveform $v_S(t) = 169 [\cos 377t] u(t)$ V.

9-47 Repeat Problem 9-45 for the input waveform $v_S(t) = 24 e^{-1000t} u(t)$ V.

9-48 Use the Laplace transformation to find the $v(t)$ that satisfies the following second-order differential equation:

$$\frac{d^2 v(t)}{dt^2} + 20 \frac{dv(t)}{dt} + 1000v(t) = 0, \quad v(0^-) = 20 \text{ V and } \frac{dv(0^-)}{dt} = 0.$$

9-49 Use the Laplace transformation to find the $v(t)$ that satisfies the following second-order differential equation:

$$\frac{d^2 v(t)}{dt^2} + 50 \frac{dv(t)}{dt} + 400v(t) = 0, \quad v(0^-) = 0 \text{ and } \frac{dv(0^-)}{dt} = 1000 \text{ V/s}$$

- 9-50 The switch in Figure P9-50 has been open for a long time and is closed at $t = 0$. The circuit parameters are $R = 500 \Omega$, $L = 2.5 \text{ H}$, $C = 2.5 \mu\text{F}$, and $V_A = 1000 \text{ V}$.
- (a) Find the circuit differential equation in $i_L(t)$ and the initial conditions $i_L(0)$ and $v_C(0)$.
- (b) Use Laplace transforms to solve for $i_L(t)$ for $t \geq 0$.

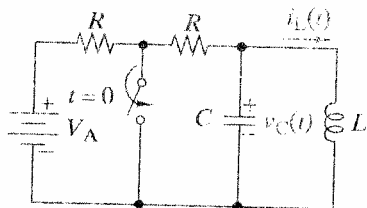


FIGURE P9-50

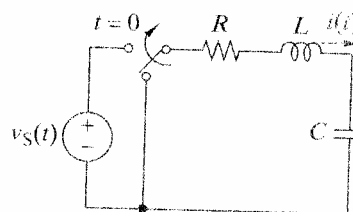


FIGURE P9-54

- 9-55 Repeat problem 9-54 when an exponential source, $v_S(t) = 10(1 - e^{-500t}) \text{ V}$, is suddenly connected to the circuit at $t = 0$.

- 9-56 Find $v_C(t)$ for $t \geq 0$ when the input to the RC circuit shown in Figure P9-56 is $v_S(t) = V_A r(t) \text{ V}$. Assume $v_C(0^-) = 0 \text{ V}$.

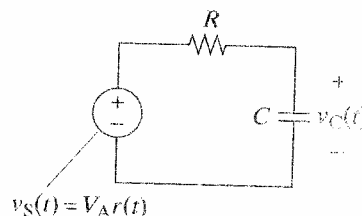


FIGURE P9-56

- 9-51 The switch in Figure P9-50 has been open for a long time and is closed at $t = 0$. The circuit parameters are $R = 500 \Omega$, $L = 2.5 \text{ H}$, $C = 2.5 \mu\text{F}$, and $V_A = 50 \text{ V}$.
- (a) Find the circuit differential equation in $v_C(t)$ and the initial conditions $i_L(0)$ and $v_C(0)$.
- (b) Use Laplace transforms to solve for $v_C(t)$ for $t \geq 0$.

- 9-52 The switch in Figure P9-52 has been closed for a long time and is opened at $t = 0$.

- (a) Find the circuit differential equation in $v_C(t)$ and the initial conditions $i_L(0)$ and $v_C(0)$.
- (b) The circuit parameters are $L = 50 \text{ H}$, $C = 0.25 \mu\text{F}$, $R_1 = 10 \text{ k}\Omega$, $R_2 = 20 \text{ k}\Omega$, and $v_S = 10u(t) \text{ V}$. Use Laplace transforms and MATLAB to solve for $v_C(t)$ for $t \geq 0$.

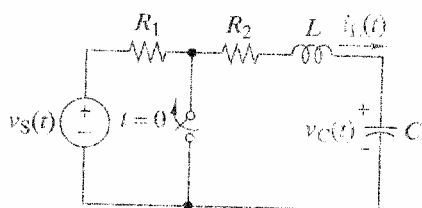


FIGURE P9-52

- 9-53 9-53 The switch in Figure P9-52 has been open for a long time and is closed at $t = 0$.

- (a) Find the circuit differential equation in $i_L(t)$ and the initial conditions $i_L(0)$ and $v_C(0)$.
- (b) The circuit parameters are $L = 50 \text{ H}$, $C = 0.25 \mu\text{F}$, $R_1 = 10 \text{ k}\Omega$, $R_2 = 10 \text{ k}\Omega$, and $v_S = 10u(t) \text{ V}$. Use Laplace transforms and MATLAB to solve for $i_L(t)$ for $t \geq 0$.

- 9-54 9-54 The RLC circuit in Figure P9-54 is in the zero state when at $t = 0$ an exponential source, $v_S(t) = V_A e^{-\alpha t} \text{ V}$, is suddenly connected to it

- (a) Find the circuit integrodifferential equation that describes the behavior of the current in the circuit.

- (b) If $R = 100 \Omega$, $L = 50 \text{ mH}$, $C = 10 \mu\text{F}$, $V_A = 10 \text{ V}$, and $\alpha = 500 \text{ s}^{-1}$, use Laplace transforms and MATLAB to solve for $i(t)$ for $t \geq 0$.

OBJECTIVE 9-4 INITIAL AND FINAL VALUE PROPERTIES (SECT. 9-6)

Given the Laplace transform of a signal, find the initial and final values of the signal waveform. See Example 9-20 and Exercise 9-31.

- 9-57 Use the initial and final value properties to find the initial and final values of the waveforms corresponding to the transforms below. If either property is not applicable, explain why.

(a) $F_1(s) = \frac{16(s+1)}{(s+2)(s+3)}$

(b) $F_2(s) = \frac{s-100}{s(s+50)}$

- 9-58 Use the initial and final value properties to find the initial and final values of the waveforms corresponding to the transforms below. If either property is not applicable, explain why.

(a) $F_1(s) = \frac{8s^2}{(s+2)(s^2+12s+13)}$

(b) $F_2(s) = \frac{s+1000}{s(s+50)(s+100)}$

- 9-59 Use the initial and final value properties to find the initial and final values of the waveforms corresponding to the transforms below. If either property is not applicable, explain why.

$$(a) F_1(s) = 400 \frac{s}{(s+10)^2 + 20^2}$$

$$(b) F_2(s) = \frac{30(3s^4 + 10s^2 + 4)}{s(s^2 + 1)(s^2 + 4)}$$

9-60 Use the initial and final value properties to find the initial and final values of the waveforms corresponding to the transforms below. If either property is not applicable, explain why.

$$(a) F_1(s) = \frac{50s(s^2 + 5s + 6)}{(s+2)(s+6)(s+12)}$$

$$(b) F_2(s) = \frac{25(s^2 + 10s + 40)}{s(s^2 - 625)}$$

9-61 Use the initial and final value properties to find the initial and final values of the waveforms corresponding to the following transforms. If either property is not applicable, explain why.

$$(a) F_1(s) = \frac{s(s+6)}{s^2 + 6s + 9}$$

$$(b) F_2(s) = \frac{20(s^2 + 10s + 100)}{s(s^2 + 20s + 100)}$$

9-62 Use the initial and final value properties to find the initial and final values of the waveform corresponding to the following transform. If either property is not applicable, explain why.

$$F(s) = \frac{80(s^3 + 3s^2 + s + 2)}{s(s^3 + 4s^2 + 4s + 16)}$$

9-63 The MATLAB function `limit` can be used to take the limit of a symbolic expression. Use MATLAB and the initial and final value properties to find the initial and final values of the waveforms corresponding to the following transforms. If either property is not applicable, explain why. Use MATLAB again to compute the waveforms corresponding to each transform and then take limits in the time domain to verify the answers found using the initial and final value properties.

$$(a) F_1(s) = \frac{(s+200)^2}{(s+20)^2(s+100)}$$

$$(b) F_2(s) = \frac{(s+20)^2}{(s+100)^2(s+200)}$$

INTEGRATING PROBLEMS

9-64 The Dominant Pole Approximation

When a transform $F(s)$ has widely separated poles, then those closest to the j -axis tend to dominate the response because they have less damping. An approximation to the waveform can be obtained by ignoring the contributions of all except the dominant poles. We can ignore the other poles simply by discarding their terms in the partial fraction expansion of $F(s)$. The purpose of this example is to examine a dominant pole approximation of the transform

$$F(s) = 10^6 \frac{s + 4000}{(s + 10000)[(s + 25)^2 + 100^2]}$$

(a) Construct a partial-fraction expansion of $F(s)$ and find $f(t)$.

(b) Construct a pole-zero diagram of $F(s)$ and identify the dominant poles.

(c) Construct a dominant pole approximation $g(t)$ by discarding the other poles in the partial fraction expansion in part (a).

(d) Plot $f(t)$ and $g(t)$ and comment on the accuracy of the approximation.

9-65 First-Order Circuit Step Response

In Chapter 7 we found that the step response of a first-order circuit can be written as

$$f(t) = f(\infty) + [f(0) - f(\infty)]e^{-t/T_c}$$

where $f(0)$ is the initial value, $f(\infty)$ is the final value, and T_c is the time constant. Show that the corresponding transform has the form

$$F(s) = K \left[\frac{s + \gamma}{s(s + \alpha)} \right]$$

and relate the time-domain parameters $f(0)$, $f(\infty)$, and T_c to the s -domain parameters K , γ , and α .

9-66 Inverse Transform for Complex Poles

In Section 9-4 we learned that complex poles occur in conjugate pairs and that for simple poles the partial fraction expansion of $F(s)$ will contain two terms of the form

$$F(s) = \dots \frac{k}{s + \alpha - j\beta} + \frac{k}{s + \alpha + j\beta} + \dots$$

Show that when the complex conjugate residues are written in rectangular form as

$$k = a + jb \text{ and } k^* = a - jb$$

the corresponding term in the waveform $f(t)$ is

$$f(t) = \dots + 2e^{-at}[a \cos(\beta t) - b \sin(\beta t)] + \dots$$

9-67 Solving State Variable Equations

With zero input, a series RLC circuit can be described by the following coupled first-order equations in the inductor current $i_L(t)$ and capacitor voltage $v_C(t)$.

$$\begin{aligned} \frac{dv_C(t)}{dt} &= \frac{1}{C} i_L(t) \\ \frac{di_L(t)}{dt} &= -\frac{1}{L} v_C(t) - \frac{R}{L} i_L(t) \end{aligned}$$

(a) Transform these equations into the s domain and solve for the transforms $I_L(s)$ and $V_C(s)$ in terms of the initial conditions $i_L(0) = I_0$ and $v_C(0) = V_0$.

(b) Find $i_L(t)$ and $v_C(t)$ for $R = 1 \text{ k}\Omega$, $L = 1 \text{ H}$, $C = 1 \text{ }\mu\text{F}$, $I_0 = 15 \text{ mA}$ and $V_0 = 10 \text{ V}$.

9-68 Complex Differentiation Property

The complex differentiation property of the Laplace transformation states that

$$\text{If } \mathcal{L}\{f(t)\} = F(s) \text{ then } \mathcal{L}\{tf(t)\} = -\frac{d}{ds}F(s)$$

Use this property to find the Laplace transforms of $f(t) = \{tg(t)\}u(t)$ when $g(t) = e^{-at}$. Repeat for $g(t) = \sin \beta t$ and $g(t) = \cos \beta t$.

9-69 Butterworth Poles

Steven Butterworth, a British engineer, 1885-1958, discovered a method of designing electric filters. He was quoted saying "An ideal electrical filter should not only completely reject the unwanted frequencies but should also have uniform sensitivity for the wanted frequencies." His algorithms are widely used in filter design, as we will see in Chapter 14. He based his design on locating the poles of his filters in a unique pattern around a circle of radius ω_C . The number of poles on the circle constitutes the order of the filter. The more poles, the better the filter. Odd-order filters include one real pole at $-\omega_C$ and pairs of complex-conjugate poles placed on a circle of radius ω_C at equal angular spacing. Even-order filters locate complex-conjugate poles placed on a circle of radius ω_C at equal angular spacing. Figure P9-69 shows the location of the poles in a third- and in a fourth-order Butterworth filter. Assuming an ω_C of 1 rad/s, what is the denominator of the Laplace transform, $F(s)$, associated with each of the two Butterworth filters shown in the figure?

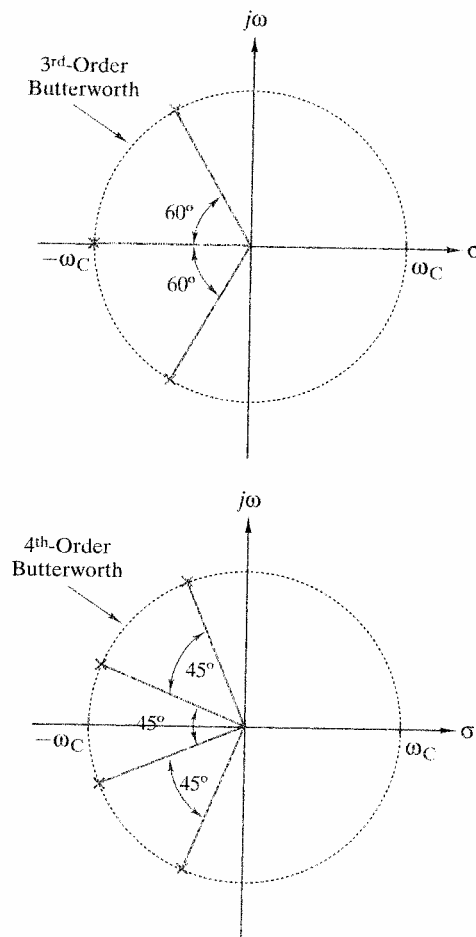


FIGURE P9-69

CHAPTER 10 s -DOMAIN CIRCUIT ANALYSIS

The resistance operator Z is a function of the electrical constants of the circuit components and of d/dt , the operator of time-differentiation, which will in the following be denoted by p simply.

Oliver Heaviside, 1887,
British Engineer

Some History Behind This Chapter

The use of operational methods to study electric circuits was pioneered by Oliver Heaviside (1850–1925). The quotation given here was taken from his book *Electrical Papers*, originally published in 1887. His resistance operator Z , which he later called impedance, is a central theme for much of electrical engineering. Heaviside does not often receive the recognition he deserves, in part because his intuitive approach was not accepted by most Victorian scientists of his day. Mathematical justification for his methods was eventually supplied by John Bromwich and others. However, no important errors were found in Heaviside's results.

Why This Chapter Is Important Today

In this chapter we use Laplace transforms to make analyzing dynamic circuits no more difficult than dc circuits. You will see that all the circuit analysis tools learned in Chapters 1 through 4 can be extended to the study of the transient response of linear circuits. In addition, pole-zero diagrams will give you a new way to visualize and predict circuit behavior.

Chapter Sections

- 10-1 Transformed Circuits
- 10-2 Basic Circuit Analysis in the s Domain
- 10-3 Circuit Theorems in the s Domain
- 10-4 Node-Voltage Analysis in the s Domain
- 10-5 Mesh-Current Analysis in the s Domain
- 10-6 Summary of s -Domain Circuit Analysis

Chapter Learning Objectives

10-1 Equivalent Impedance (Sects. 10-1, 10-2)

Given a linear circuit, use series and parallel equivalence to find the equivalent impedance at specified terminal pairs. Select element values to obtain specified pole locations.

10-2 Basic Circuit Analysis Techniques (Sects. 10-2, 10-3)

Given a linear circuit:

- (a) Determine the initial conditions (if not given) and transform the circuit into the s domain.
- (b) Solve for zero-state and zero-input responses using circuit reduction, the unit output method, Thévenin or Norton equivalent circuits, or superposition.
- (c) Identify the forced and natural poles in the responses, or select circuit parameters to place the natural poles at specified locations.

10-3 General Circuit Analysis (Sects. 10-4, 10-5, 10-6)

Given a linear circuit:

- (a) Determine the initial conditions (if not given) and transform the circuit into the s domain.
- (b) Solve for zero-state and zero-input response transforms and waveforms using node-voltage or mesh-current methods.
- (c) Identify the forced and natural poles in the responses or select circuit parameters to place the natural poles at specified locations.